Homework

You must complete problems 1 through 3 and then choose 1 of either problem 4 or 5. If you complete both 4 and 5, you will get a max of 5 points added onto your exams (based on problem correctness).

All text responses must be written by hand (problems 1 through 3); you may use electronic mediums such as a tablet or write them on paper and take a picture of it. All writing must be neat, organized, and legible. Workflow section problems will require a plot from software such as excel or python, and your code must be uploaded separately. Submissions will be uploaded to Blackboard. You may work with others, but your final work must be your own.

Section 1: Theoretical Concepts

Question 1 (CA): Explain the purpose of interatomic potentials in MD. I would then like you to find 3 different types of interatomic potentials than what was discussed in class and do the following: (a) write and label their mathematical form to predict the total potential energy, (b) determine what types of materials the potential is designed for (ex: metals) and explain why, and (c) explain the potential pitfalls of using this potential to run MD.

Question 2 (CA): Draw and label a flowchart for a typical MD algorithm workflow starting from making a structure through calculating properties. Explain the purpose of each component within the MD framework and their potential pitfalls if used incorrectly.

Question 3 (CA, CC): For this problem I would like you to: (a) define the mathematical forms of each statistical ensemble's Hamiltonian (ex: NVT), define all terms, and define what each "block" in the function represents physically, (b) come up with one example of a materials science problem that would require each of these ensembles to simulate correctly and explain why that ensemble should be used to simulate that problem. You cannot use the examples we discussed in class during any lecture.

Section 2: Problem Solving

Question 4 (CA, CB):

Objective. Use molecular dynamics to determine how the equilibrium lattice parameter of FCC aluminum changes with temperature. You will use the NPT ensemble so that the simulation cell can change its volume while the pressure and temperature are controlled, then use the equilibrated MD data to estimate a linear coefficient of thermal expansion.

Supplied files. You will be given the LAMMPS input script, an Al EAM/alloy potential, submission script, and a Python analysis template. The LAMMPS input creates the FCC Al structure internally; no starting structure file is required.

in.thermal_expansion builds a 4 × 4 × 4 FCC Al supercell and runs NPT MD at 100, 300, 500, and 700 K.

analyze_thermal_expansion_template.py reads the production data and prepares the lattice-parameter data for analysis.

Instructor-provided Al EAM/alloy potential file.

A. Run the NPT simulations

Before submitting the job, inspect the LAMMPS input and identify the commands that create the FCC structure, assign atomic velocities, set the timestep, and define the NPT thermostat/barostat. Run the calculation on the SCC using the course submission workflow.

The input performs an equilibration period followed by a production period at each temperature. Confirm that the following production files are generated:

thermal_100.dat
thermal_300.dat
thermal_500.dat
thermal_700.dat

Do not use a single final MD frame to determine the lattice parameter. The simulation cell fluctuates continuously at finite temperature, so the relevant quantity is a time average over the equilibrated production trajectory.

B. Examine equilibration and fluctuations

Inspect the LAMMPS thermodynamic output for at least two temperatures. Confirm that the temperature and pressure fluctuate around their target values and that the cell dimensions and volume fluctuate during the NPT trajectory.

In 2–4 sentences, explain why the instantaneous pressure can fluctuate substantially even though the target pressure is 0 bar, and why these fluctuations do not necessarily indicate that the simulation has failed to equilibrate.

C. Determine the lattice parameter

Run the supplied analysis template from the directory containing the four thermal_*.dat files:

python analyze_thermal_expansion_template.py

The supplied code reads the LAMMPS files, calculates an effective FCC lattice parameter from each instantaneous cell volume, and computes the average lattice parameter and uncertainty at each temperature. For an FCC crystal containing N atoms:

$a(T) = [4\ V(T) / N]^{(1/3)}$

Complete the plotting section of the Python template. Plot average lattice parameter versus target temperature and include error bars using the supplied uncertainty. Label both axes with units and save the figure as:

al_thermal_expansion.png

D. Calculate the coefficient of thermal expansion

Approximate the temperature dependence over the simulated range with a linear model:

$a(T) = a_{ref} + mT$

where $m = da/dT$. Fit the data, add the fitted line to your plot, and estimate the linear coefficient of thermal expansion using:

$\alpha = (1 / a_{ref})\ (da/dT)$

Report the fitted slope da/dT in Å/K.

Report the reference lattice parameter a_{ref} in Å.

Report α in K^{-1}.

E. Explain the ensemble choice

In a short response, explain why NPT is the appropriate ensemble for this property. Explicitly contrast it with NVT, where the simulation-cell volume is fixed. Your response should connect the controlled ensemble variables to the physical quantity being measured.

F. Interpretation

Write a 150–250 word interpretation of your results. Address all of the following points:

Does the average lattice parameter increase monotonically with temperature?

Why is time averaging required for an MD-derived materials property?

Why does NPT permit thermal expansion to occur naturally?

What would physically happen if the same heating study were performed in NVT?

What is one limitation of fitting a single linear thermal-expansion coefficient over the entire 100–700 K interval?

What to submit

Item	**Required submission**
1	Completed analyze_thermal_expansion_template.py.
2	al_thermal_expansion.png with averaged data, error bars, and your linear fit.
3	thermal_expansion_summary.csv generated by the supplied script.
4	A table listing the average lattice parameter at all four temperatures and your fitted α.
5	Your short response to Part E.
6	Your 150–250 word interpretation from Part F.

Question 5 (CA, CB, CC):

Objective. Fit a simple Lennard-Jones (LJ) potential to fictitious Al–Al reference data, use that fitted potential as an ASE calculator for an Al_2 geometry optimization, and compare the optimized bond length with a first-principles VASP calculation. The goal is to connect parameter fitting, atomistic optimization, and independent model validation in one workflow.

The Lennard-Jones potential is

$U(r) = 4 \varepsilon [(\sigma/r)^{12} - (\sigma/r)^{6}]$

This potential is intentionally simple. It should not be interpreted as a generally accurate potential for metallic aluminum; that limitation is part of the point of the problem.

Supplied files. You will receive fictitious Al–Al energy data, a partially completed ASE/Python workflow, VASP INCAR, KPOINTS, and POTCAR files, and a complete comparison script.

data/al_dimer_fictitious.csv: fictitious Al–Al energy versus separation data.

fit_lj_and_optimize_ase_template.py: performs the LJ fit but requires you to create the starting dimer and implement the ASE optimizer.

vasp/INCAR, vasp/KPOINTS, and vasp/POTCAR are supplied VASP inputs.

compare_lj_vasp.py: complete analysis script used after both the ASE and VASP calculations are finished.

Not supplied. You must create the VASP POSCAR and submit.sh yourself.

A. Fit the Lennard-Jones potential

Run the supplied Python script:

python fit_lj_and_optimize_ase_template.py

The script reads the fictitious reference data and fits ε and σ. It also calculates the analytical LJ equilibrium separation:

$r_{eq} = 2^{(1/6)} \sigma$

The script saves the fitted potential curve and fit parameters before stopping at the first student TODO. Report the fitted ε, fitted σ, fit RMSE, and analytical equilibrium separation. Inspect results/lj_fit.png and briefly assess whether the LJ function reproduces the supplied reference data.

B. Construct the Al_2 dimer in ASE

Complete the first TODO section of fit_lj_and_optimize_ase_template.py. Construct an ASE Atoms object for Al_2 satisfying all of the following requirements:

20 Å cubic simulation cell.

Initial Al–Al separation of 3.50 Å.

Dimer centered well away from the box boundaries with pbc=False.

You must determine the Cartesian coordinates of the two atoms yourself. Do not modify the required variable name used by the supplied script.

C. Geometry optimize with the fitted potential

The supplied script automatically attaches the fitted LJ parameters to an ASE LennardJones calculator. Complete the second TODO section by implementing an ASE geometry optimization.

Choose an ASE geometry-optimization algorithm.

Save the trajectory as results/Al2_LJ_optimization.traj.

Save the optimizer log as results/Al2_LJ_optimization.log.

Converge to a maximum force of no more than 0.01 eV/Å.

After the optimization, the supplied code will save the final Al–Al distance and optimized structure. Compare the ASE result with the analytical LJ minimum and explain why the two values should be nearly identical if the fit and geometry optimization are implemented correctly.

D. Geometry optimize Al_2 with VASP

Create your own VASP POSCAR for the same starting geometry used in ASE:

20 Å cubic cell.

Initial Al–Al separation of 3.50 Å.

Dimer centered within the simulation cell.

Use the supplied INCAR, KPOINTS, and POTCAR for Al, and create or reuse an appropriate SCC submit.sh. Run the VASP geometry optimization and confirm that both CONTCAR and OUTCAR are generated successfully.

E. Compare the fitted potential with VASP

The supplied compare_lj_vasp.py script is complete; you should not need to modify it. However, it expects your ASE and VASP outputs to remain in the directory structure used in the distributed problem. In particular, the script expects:

results/lj_fit_parameters.json
results/lj_optimization_results.json
vasp/CONTCAR
vasp/OUTCAR

Once both calculations are complete, run:

python compare_lj_vasp.py

The script compares the analytical LJ minimum, the ASE + fitted-LJ optimized distance, and the VASP/PBE optimized distance. It also calculates the ASE-LJ versus VASP difference and percent difference, and generates VASP energy/force convergence plots.

Do not directly compare the absolute LJ energy with the absolute VASP total energy. These two methods do not share the same energy zero.

F. Interpretation

Write a 150–250 word interpretation discussing the relationship between the fictitious training data, fitted LJ potential, ASE optimization, and VASP result. Address all of the following:

Why should the ASE-LJ optimized geometry agree closely with the analytical LJ minimum?

Why can the fitted LJ geometry disagree with VASP even when the LJ fit itself is excellent?

What important physics is missing from a simple two-body Lennard-Jones description of metallic Al?

What does this exercise illustrate about validating an interatomic potential outside the fitting procedure itself?

What to submit

Item	Required submission
1	Completed fit_lj_and_optimize_ase_template.py.
2	results/lj_fit.png.
3	results/lj_fit_parameters.json.
4	results/lj_optimization_results.json.
5	Your VASP POSCAR.
6	Your VASP submit.sh.
7	results/lj_vs_vasp_comparison.csv.
8	results/lj_vs_vasp_comparison.png.
9	results/vasp_energy_convergence.png.
10	results/vasp_force_convergence.png.
11	Your 150–250 word interpretation from Part F.